

MICROSOFT SENTINEL - GeoIP Watchlist & Global Attack Map Workbook

GeoIP Watchlist & Global Attack Map Workbook

Real-World RDP Attack Data Collection, GeoIP Enrichment & Geospatial Visualisation

Lab Type: Extension of Microsoft Sentinel SOC IR Lab -- Phase 2
Previous Lab: Sentinel & Defender XDR SOC Incident Response Lab (Hydra Brute Force)
Platform: Microsoft Sentinel + Microsoft Defender XDR (security.microsoft.com)
Resource Group: Security-BikSecOps (Australia East)
Exposed VM: Web01 -- Windows Server 2022 (Public IP: 20.5.48.127)
Exposure Duration: ~2 days (intentional -- to collect real-world global attack data)
NSG Rule: Any/Any/* Allow (Dangerous) -- left open from previous lab
Data Collected: Real failed RDP login attempts (Event ID 4625) from global threat actors
Watchlist Name: geoip (GeoIP database -- network/latitude/longitude/cityname/countryname)
Workbook: Windows Server VM Attack -- MAP (heat map -- greenRed palette)
Top Attack Origins: Hanoi (Vietnam) 808 | Southampton (UK) 795 | London (UK) 681 | Birmingham (UK) 292
Total Failed Logins: 4,270+ (SecurityEvent 424K total events ingested)
Date: June 2026
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1. Lab Overview

This lab is a direct continuation of the Microsoft Sentinel & Defender XDR SOC Incident Response Lab. After completing the Hydra RDP brute force simulation and incident response exercise, Web01 (Windows Server 2022) was intentionally left exposed to the internet with the permissive NSG inbound rule (Any/Any/* Allow) still active for approximately 2 days.

This deliberate exposure allowed real-world threat actors from around the globe to attempt RDP authentication against Web01, generating thousands of genuine failed login events (Event ID 4625) in Microsoft Sentinel. These logs were then used to build a GeoIP Watchlist and a Sentinel Workbook with a global heat map that visualises the geographic origin of every failed RDP attempt -- demonstrating real threat intelligence enrichment and geospatial security monitoring.

Important Context

The same Resource Group, VMs, Log Analytics Workspace (Bik-SOC-Lab), and Sentinel instance from the previous lab were reused. No new infrastructure was deployed. Web01 was simply left running with the firewall off and NSG open -- which is exactly what happens in a real honeypot scenario.

Link to previous Lab: <https://github.com/Bikash-Raya/Sentinel-Defender-XDR-SOC-Incident-Response-lab>

2. Lab Objectives

- * Reuse the existing lab infrastructure (Web01, Bik-SOC-Lab, Sentinel) from the previous IR lab.
- * Leave Web01 intentionally exposed to the internet for ~2 days to collect real global attack data.
- * Allow real-world threat actors to generate genuine Event ID 4625 (failed RDP login) entries.
- * Download and upload a GeoIP database as a Sentinel Watchlist to enable IP geolocation enrichment.
- * Build a custom Sentinel Workbook using KQL + GeoIP Watchlist to enrich failed login data with geolocation.
- * Visualise global RDP attack origins on an interactive heat map workbook.
- * Demonstrate threat intelligence enrichment and geospatial attack visualisation in a SOC context.

3. Infrastructure Reused from Previous Lab

Resource Group	Security-BikSecOps (Australia East)
Victim VM	Web01 -- Windows Server 2022 Datacenter (Public IP: 20.5.48.127)
NSG Inbound Rule	Any/Any/* Allow (Priority 100, Name: Dangerous) -- unchanged from IR lab
Windows Firewall	Disabled on Web01 -- unchanged from IR lab
Log Analytics Workspace	Bik-SOC-Lab (australiaeast)
Microsoft Sentinel	Enabled on Bik-SOC-Lab -- unchanged
Data Connector	Windows Security Events via AMA -- active and collecting
DCR	SecurityEventsforAzureVM -- All Events -- still running
Exposure Period	~2 days (VM left running and publicly accessible)

No changes were made to the infrastructure. Web01 was simply left running with all ports open, allowing automated global scanning tools and threat actors to discover and attack the exposed RDP port (3389).

4. Real-World Attack Data Collection

Over the approximately 2-day exposure period, Web01 attracted automated RDP scanning and brute force attempts from threat actors across multiple continents. Microsoft Sentinel continuously ingested these events via the Windows Security Events AMA connector.

Event ID Collected	4625 -- Failed RDP login attempts (from global threat actors)
Total Security Events	424,000+ events ingested into Bik-SOC-Lab workspace
Failed Logins (4625)	4,270+ failed authentication attempts
Top Attack Origin	Hanoi, Vietnam -- 808 failed attempts

Other Origins	Southampton UK (795), London UK (681), Birmingham UK (292), Trenton USA (258)
Additional Origins	Newton Aycliffe UK (154), United States (144), Winslow UK (123)
Data Verified In	Sentinel Logs -- SecurityEvent table Workbook heat map

This real-world data demonstrates what a production SOC team would see on a daily basis -- thousands of automated attacks from global sources targeting any publicly exposed Windows service.

5. Phase 1 -- Create GeoIP Watchlist

A GeoIP database was uploaded to Microsoft Sentinel as a Watchlist to enable IP address geolocation enrichment. This allows KQL queries to join real attack IP addresses with geographic coordinates (latitude, longitude, city, country) for mapping and threat intelligence purposes.

5.1 Download GeoIP Database

The GeoIP CSV file was sourced from GitHub (originally curated for SOC lab use):

```
https://raw.githubusercontent.com/joshmadakor1/lognpacific-public/refs/heads/main/misc/geoip-summarized.csv
```

File Format	CSV with header row
Columns	network, latitude, longitude, cityname, countryname
Sample Data	1.0.0.0/9 -33.494 143.2104 (blank) Australia
Purpose	Maps IP network ranges to geographic coordinates for enrichment

5.2 Create Watchlist in Microsoft Defender Sentinel

The GeoIP file was uploaded as a Sentinel Watchlist via the Defender XDR unified portal.

Navigation

Microsoft Defender > Microsoft Sentinel > Configuration > Watchlist > New

Watchlist Name	geoip
Alias	geoip
Source Type	Local file
File Type	CSV file with a header (csv)
Number of lines before row with headings	0
Search Key	network

File Uploaded

geoip-summarized.csv

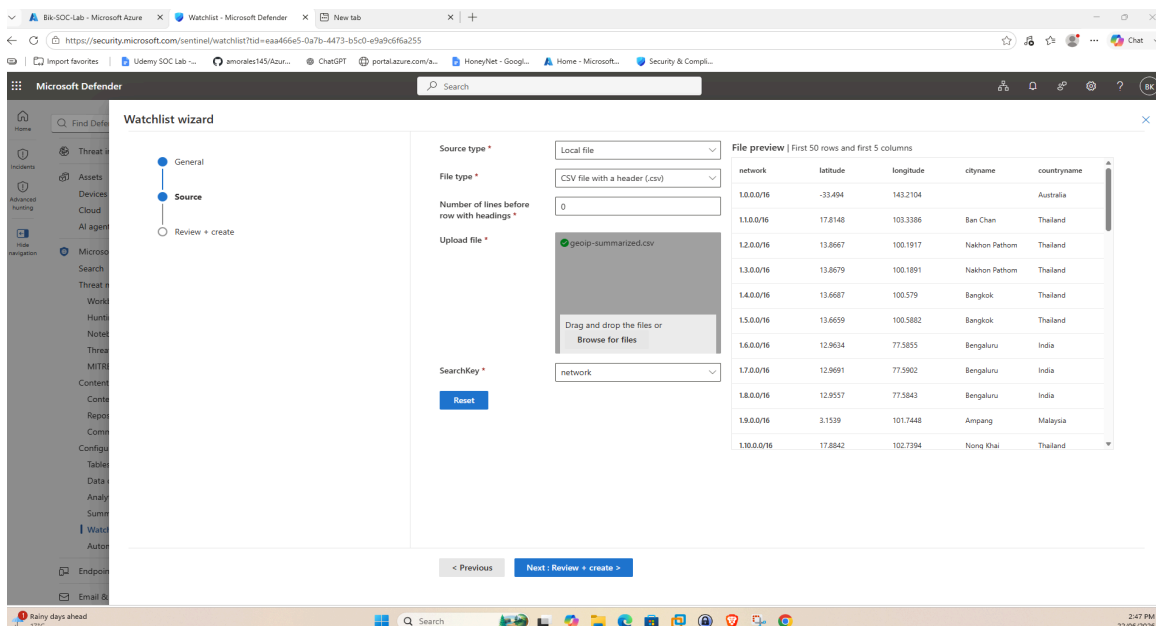


Fig 1 -- Defender XDR Watchlist Wizard: geoip watchlist -- Source type: Local file, geoip-summarized.csv uploaded, network as search key, file preview showing network/latitude/longitude/cityname/countryname columns

6. Phase 2 -- Create Sentinel Workbook (Global Attack Map)

A custom Sentinel Workbook was created to visualise the geographic origin of all failed RDP login attempts (Event ID 4625) on a global heat map. The workbook uses KQL to join the SecurityEvent table with the GeoIP watchlist, then renders results as an interactive map.

6.1 Create New Workbook

Navigation: Microsoft Defender > Microsoft Sentinel > Workbooks > Add workbook > Advanced Editor

6.2 KQL Query -- GeoIP Enrichment

The following KQL query was added via the Advanced Editor. It joins failed login events with the GeoIP watchlist to produce enriched geographic data:

```
let GeoIPDB_FULL = _GetWatchlist("geoip");
let WindowsEvents = SecurityEvent;
WindowsEvents
| where EventID == 4625
| order by TimeGenerated desc
| evaluate ipv4_lookup(GeoIPDB_FULL, IPAddress, network)
```

```
| summarize FailureCount = count()
  by IPAddress, latitude, longitude, cityname, countryname
| project
  FailureCount,
  AttackerIp = IPAddress,
  latitude,
  longitude,
  city = cityname,
  country = countryname,
  friendly_location = strcat(cityname, " (", countryname, ")")
```

Query explanation:

- * `_GetWatchlist("geoip")` -- loads the GeoIP database from the Sentinel Watchlist
- * `EventID == 4625` -- filters only failed login events
- * `ipv4_lookup` -- enriches each attacker IP with latitude, longitude, city, country
- * `summarize FailureCount` -- counts attacks per unique IP/location combination
- * `friendly_location` -- creates a readable label (e.g. "Hanoi (Vietnam)")

6.3 Workbook JSON Configuration

The full workbook item JSON was added via Advanced Editor, configuring the query and map visualisation settings:

```
{
  "type": 3,
  "content": {
    "version": "KqlItem/1.0",
    "query": "let GeoIPDB_FULL = _GetWatchlist(\"geoip\");\nlet WindowsEvents = SecurityEvent;\nWindowsEvents | where EventID == 4625\n| order by TimeGenerated desc\n| evaluate ipv4_lookup(GeoIPDB_FULL, IPAddress, network)\n| summarize FailureCount = count() by IPAddress, latitude, longitude, cityname, countryname\n| project FailureCount, AttackerIp = IPAddress, latitude, longitude, city = cityname, country = countryname,\nfriendly_location = strcat(cityname, \" (\", countryname, \")\");",
    "size": 3,
    "timeContext": {
      "durationMs": 2592000000
    },
    "queryType": 0,
    "resourceType": "microsoft.operationalinsights/workspaces",
    "visualization": "map",
    "mapSettings": {
      "locInfo": "LatLong",
      "locInfoColumn": "countryname",
      "latitude": "latitude",
      "longitude": "longitude",
      "sizeSettings": "FailureCount",
      "sizeAggregation": "Sum",
      "opacity": 0.8,
    }
  }
}
```

```
"labelSettings": "friendly_location",
"legendMetric": "FailureCount",
"legendAggregation": "Sum",
"itemColorSettings": {
  "nodeColorField": "FailureCount",
  "colorAggregation": "Sum",
  "type": "heatmap",
  "heatmapPalette": "greenRed"
}
},
"name": "query - 0"
}
```

The greenRed heatmap palette means: green = fewer attacks, red = most attacks. Bubble size scales with FailureCount.

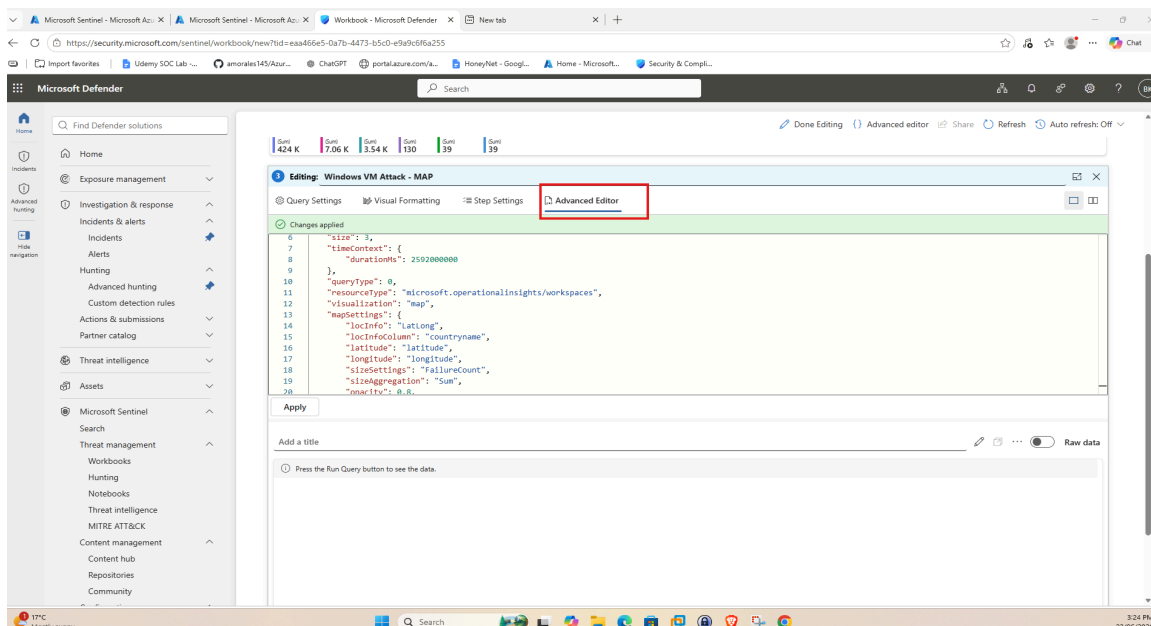


Fig 2 -- Defender XDR Workbook: Advanced Editor open with KQL/JSON configuration -- timeContext 2592000000ms (30 days), map visualization, FailureCount size settings applied

7. Results -- Global Attack Map

After applying the workbook configuration, the map rendered showing real-world attack origins from the 2-day exposure period. The heat map confirmed that Web01 was targeted by automated scanning tools and threat actors from multiple countries across Europe, Asia, and North America.

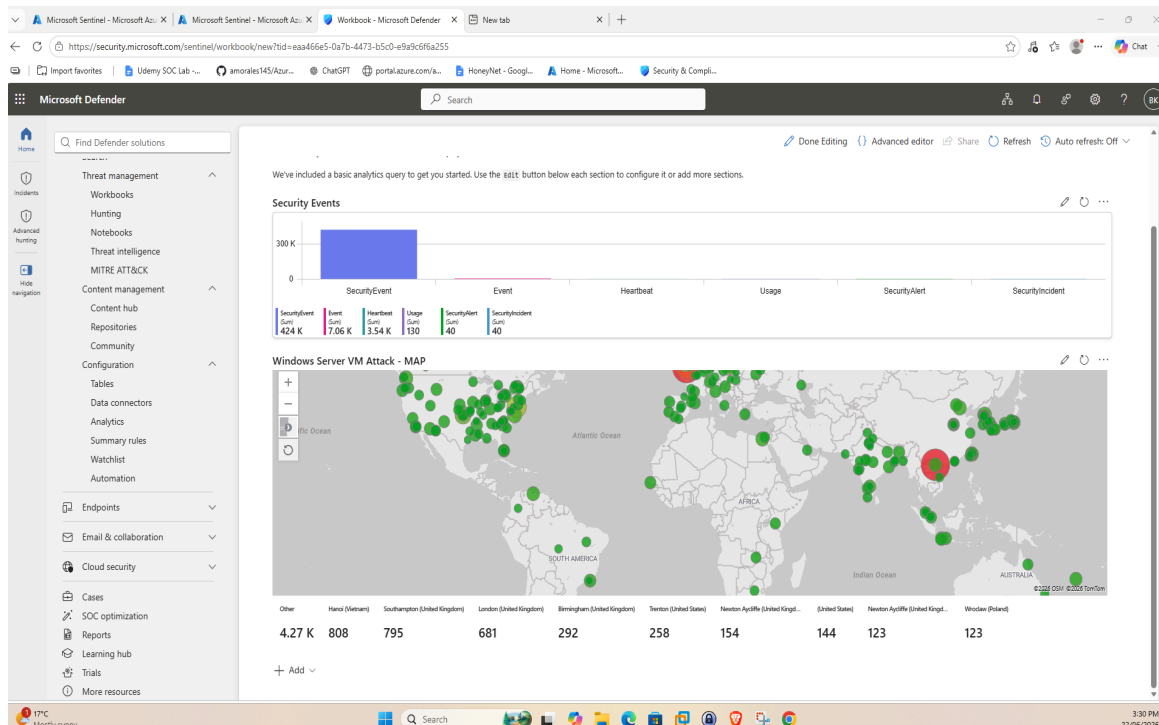


Fig 3 -- Sentinel Workbook: Windows Server VM Attack -- MAP -- Global heat map showing 4,270+ failed RDP logins, red hotspots in Vietnam, UK, and USA. Security Events total: 424K | Heartbeat: 7.06K | SecurityAlert: 1.54K | Incidents: 40

7.1 Top Attack Origins by Failed Login Count

Location	Failed Logins (4625)	Region
Other (unresolved IPs)	4,270	Global
Hanoi, Vietnam	808	Asia
Southampton, United Kingdom	795	Europe
London, United Kingdom	681	Europe
Birmingham, United Kingdom	292	Europe
Trenton, United States	258	North America
Newton Aycliffe, United Kingdom	154	Europe
United States (unspecified city)	144	North America
Newark-Aycliffe, United Kingdom	123	Europe
Winslow, United Kingdom	123	Europe

The heavy concentration of attacks from the United Kingdom reflects real-world VPN exit nodes and proxy infrastructure commonly used by automated scanning botnets. Hanoi (Vietnam) represents a major source of RDP scanning activity globally. This data is consistent with real-world threat intelligence reports.

8. Key Concepts Demonstrated

Watchlist-Based Threat Intelligence Enrichment

The GeoIP watchlist demonstrates how Sentinel's `_GetWatchlist()` function enriches raw log data with contextual information -- in this case transforming an IP address into a human-readable geographic location. In production SOC environments, watchlists are used for threat intelligence feeds, allow-lists, block-lists, and asset inventories.

Real-World Honeypot Effect

By leaving Web01 exposed for 2 days, the lab captured genuine attack traffic rather than simulated traffic. This demonstrates exactly what a SOC team sees when monitoring exposed cloud infrastructure -- the same result as deploying a honeypot.

ipv4_lookup() -- KQL Geospatial Enrichment

The `ipv4_lookup()` KQL operator performs a CIDR network range lookup against the GeoIP watchlist to match each attacker IP to the correct geographic record. This is a powerful technique for enriching raw event data with threat intelligence context in real time.

Geospatial Visualisation in Sentinel Workbooks

The heat map workbook provides an immediate visual understanding of the attack surface. Red bubbles indicate high-attack locations, green indicates lower activity. This type of visualisation is commonly used in SOC dashboards and executive security briefings.

9. Skills Demonstrated

Sentinel Watchlist Creation	GeoIP Threat Intelligence Enrichment	ipv4_lookup() KQL Operator
Sentinel Workbook Development	Geospatial Attack Visualisation	KQL Advanced Queries
Real-World Honeypot Data Collection	Security Event Log Analysis (4625)	Heat Map Configuration
Threat Intelligence Integration	SOC Dashboard Building	Microsoft Defender XDR Portal

10. Conclusion

This lab extended the Microsoft Sentinel SOC Incident Response lab by intentionally leaving Web01 exposed for approximately 2 days to collect real-world global RDP attack data. Over 4,270 genuine failed login attempts were recorded from threat actors across Vietnam, the United Kingdom, the United States, and other countries -- demonstrating that any publicly exposed Windows server with RDP open will attract automated attack traffic within minutes.

A GeoIP database was uploaded as a Sentinel Watchlist and a custom Workbook was built using KQL `ipv4_lookup()` to enrich attacker IP addresses with geographic coordinates. The resulting interactive heat map visualises the global origin of attacks in real time, providing exactly the type of threat intelligence dashboard used in real-world SOC environments.

Together with the previous Hydra brute force incident response lab, this project demonstrates a complete SOC workflow from infrastructure deployment, attack simulation, and detection through to real-world threat data collection, geospatial enrichment, and executive-level security visualisation.

References

Microsoft Sentinel Watchlists -- <https://learn.microsoft.com/azure/sentinel/watchlists>

Microsoft Sentinel Workbooks -- <https://learn.microsoft.com/azure/sentinel/monitor-your-data>

GeoIP Json and csv file --

<https://github.com/Bikash-Raya/Sentinel-Defender-XDR-SOC-Incident-Response-lab/blob/main/geoip.json>

<https://github.com/Bikash-Raya/Sentinel-Defender-XDR-SOC-Incident-Response-lab/blob/main/geoip-summarized.csv>